

Homework 1

Read. Rogers and Girolami, Sections 1.1 - 1.3

Problem 1. Rogers and Girolami, Exercise 1.3

Problem 2 (Matrix Transposes). We often denote the set of all $m \times n$ matrices with entries from $\mathbb{R}$ by $M_{m \times n}(\mathbb{R})$, or simply $M_n(\mathbb{R})$ if $m = n$. Prove that for $A \in M_{m \times k}(\mathbb{R})$ and $B \in M_{k \times n}(\mathbb{R})$, $(AB)^T = B^T A^T$.

Hint: More precisely, show that $((AB)^T)_{ij} = (B^T A^T)_{ij}$ for any i, j . Use the definitions from Comments 1.6 and 1.7 in Section 1.3.

Problem 3 (Matrix Calculus). Let's prove the identities that are taken for granted in Table 1.4... Recall that for a function $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$, we define its i 'th *partial derivative* to be the function $\frac{\partial \mathcal{L}}{\partial \theta_i} : \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$\frac{\partial \mathcal{L}}{\partial \theta_i}(\theta) = \lim_{h \rightarrow 0} \frac{\mathcal{L}(\theta + h e_i) - \mathcal{L}(\theta)}{h}$$

if the limit exists, where $e_i \in \mathbb{R}^n$ is the i 'th unit vector. Define $\mathcal{L}$'s *gradient* as the function $\nabla \mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by

$$\nabla \mathcal{L}(\theta) = \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial \theta_1}(\theta) \\ \frac{\partial \mathcal{L}}{\partial \theta_2}(\theta) \\ \vdots \\ \frac{\partial \mathcal{L}}{\partial \theta_n}(\theta) \end{bmatrix}.$$

We also sometimes notate the gradient by $\frac{\partial \mathcal{L}}{\partial \theta} : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Fix $b \in \mathbb{R}^n$ and $A \in M_n(\mathbb{R})$. Verify the following:

- (a) For $\mathcal{L}(\theta) = \theta^T \theta$, $\nabla \mathcal{L}(\theta) = 2\theta$.
- (b) For $\mathcal{L}(\theta) = \theta^T b = b^T \theta$, $\nabla \mathcal{L}(\theta) = b$
- (c) For $\mathcal{L}(\theta) = \theta^T A \theta$, $\nabla \mathcal{L}(\theta) = (A + A^T)\theta$. What if A is symmetric?

Problem 4 (Normal Equations). In this problem, we'll generalize the computations from Section 1.3 for a multiple linear regression model with d parameters. Here, we are given training inputs $x^{(i)} \in \mathbb{R}^d$ for $1 \leq i \leq N$ and a vector $y \in \mathbb{R}^N$ containing the target value $y^{(i)}$ for each input. We define the *design matrix* X to be the $N \times d$ matrix containing the training inputs in its rows:

$$X = \begin{bmatrix} -(x^{(1)})^T & - \\ -(x^{(2)})^T & - \\ \vdots & \\ -(x^{(N)})^T & - \end{bmatrix}, \quad y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(N)} \end{bmatrix}.$$

As before, we want to find a parameter vector $\theta \in \mathbb{R}^d$ to minimize the mean squared error loss function $\mathcal{L} : \mathbb{R}^d \rightarrow \mathbb{R}$ of our model:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N (\theta^T x^{(i)} - y^{(i)})^2.$$

- (a) The *euclidean norm* of a vector $a \in \mathbb{R}^n$ is defined to be $\|a\|_2 = \sqrt{\sum_{i=1}^n a_i^2}$. Use the euclidean norm to show that $\mathcal{L}(\theta) = \frac{1}{N} (X\theta - y)^T (X\theta - y)$. *Hint:* You probably don't want to expand this... Notice that $a^T a = \|a\|_2^2$.
- (b) Use part (a) and the results from Problems 2 and 3 to compute $\nabla \mathcal{L}(\theta)$. Set the result to 0 and solve for θ to obtain the generalized version of Equation 1.16, otherwise known as the *normal equations*:

$$\theta = (X^T X)^{-1} X^T y.$$